Roll No: _____

Thapar Institute of Engineering and Technology
Department of Computer Science and Engineering
Auxiliary Examination- Feb. 2026

Course Name: Quantum Computing | **Course Code: UCS619**
Time: 3 hours | Date: Feb. 23, 2026
Max Marks: 100 | Faculty: Dr. Vaibhav Pandey

Note: *Attempt any **ten** questions.*

1. Explain the difference among a qubit, a qutrit, and a qudit. What is Born's rule for measurement to find the basis states of any quantum system? **(10)**

2. Write down the unitary matrix for the following quantum gates:
   - $H^\dagger$ gate
   - $T^\dagger$ gate
   - $S^\dagger$ gate
   - $RY(\theta)$ gate
   - $RZ(\theta)$ gate

   (2*5=10)

3. Describe the following quantum postulates:
   1. State postulate
   2. Measurement postulate
   3. Evolution postulate
   4. Composition postulate

   (10)

4. What is the difference between a product state and an entangled state? Explain with an example. Moreover, draw the quantum circuit to generate any two of the Bell states. **(10)**

5. What do you mean by **Quantum Teleportation**? Explain its complete working with a proper circuit diagram. **(10)**

6. A system is in the state

$$|\phi\rangle = \frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

   (a) What is the probability that measurement finds the system in the state $|\phi\rangle = |01\rangle$?

   (b) What is the probability that measurement finds the first qubit in the state $|0\rangle$? What is the state of the system after measurement?

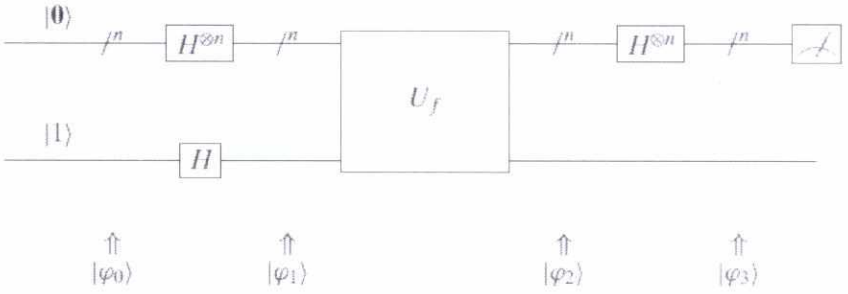
   (5+5)

7. (a) Explain how the BB84 quantum cryptography scheme secures communication.

   (b) In the B92 protocol, state the four rules based on which Bob decides his observation in the + basis or the × basis. **(5+5)**

8. Given below is the quantum circuit for the Deutsch–Jozsa algorithm. Derive the quantum states $|\phi_0\rangle$ $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$.

(10)



9. Explain the following error detection and correction schemes with circuit diagrams.

   (a) 3 qubit **bit flip**

   (b) 3 qubit **phase flip**

(10)

10. Given the search space $|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle$ (that is, we are given a function $f : \sum^3 \to \sum$ where $\sum = \{0, 1\}$). Suppose $|110\rangle$ is marked as present (in other words, $f(110) = 1$). Calculate the different steps of Grover's algorithm to find the element $|110\rangle$ in the search space. **(10)**

11. Derive the equivalent quantum circuits for the following classical Boolean circuits:

    (a) Half adder

    (b) Full adder

(10)